Laboratory 0:
An Introduction to the ECE 280 Labs!

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Note on Prompts

Throughout this and future lab documents, there will be four different prompts to let you know specific work that needs to be done.

- **Pre-lab Deliverable (N):** Work to be documented in the pre-laboratory assignment.
- **Checkpoint (N):** Work to be shown to a TA during the laboratory.
- **Discussion (N):** Questions to be discussed with your lab partner and then answered in the post-laboratory assignment.
- **Deliverable (N):** Other items to be completed individually for the post-laboratory assignment.

1 Abstract

This semester, you will interface high-level software (MATLAB) with physical embedded hardware (Arduino Mega). To ensure a smooth ”sampling pipeline” in future labs, your environment must be configured correctly. This lab guides you through the installation of MATLAB R2023b+, Arduino IDE 2.x, and the necessary hardware drivers to communicate with your Mega 2560.

2 Objectives

In this assignment, you will:

1. Locate the ECE 280 laboratory in Teer 116.
2. Verify MATLAB version compatibility and install required toolboxes.
3. Configure Arduino IDE 2.x with the correct Board Support Packages and Libraries.
4. Solve USB-to-Serial driver issues (the #1 cause of lab connectivity failures).

3 Pre-Laboratory Exercise

The ECE 280 labs meet in **Teer 116**.

- Enter Teer via the main entrance (Floor 2).
- Go down the open staircase to the Floor 1 Lobby.
- Pass the elevator and walk down the hallway past the EGR 201 lab.
- Take a right at the end of the corridor, past the Dean's portraits.

Pre-lab Deliverable (1/1): Take a selfie at the lab door and upload the image to the Gradescope assignment.

4 Laboratory Exercises

4.1 MATLAB Installation & Toolboxes

Required MATLAB version: R2020a minimum (R2023b+ strongly recommended).

Why this version floor matters:

- `exportgraphics()` was introduced in R2020a. Older versions fail figure-export deliverables.
- `serialport()`, `configureTerminator()`, and `readline()` were introduced in R2019b.
- Legacy `serial()` API is deprecated and not supported in our starter scripts.

Base MATLAB note: `fft`, `abs`, `plot`, `xlabel`, `ylabel`, `legend`, `zeros`, `ceil`, and `str2double` are included in base MATLAB.

How to verify your installation:

Table 1. MATLAB Toolboxes Required for ECE 280 Lab Sequence

Toolbox	Status	Why Needed / Typical Functions
Instrument Control Toolbox	**Required**	Serial communication with Arduino: `serialport`, `configureTerminator`, `readline`, `flush`.
Signal Processing Toolbox	**Required**	Filter/frequency analysis and report questions: `freqz`, `fir1`, `db2mag`.
DSP System Toolbox	**Required**	Streaming/spectrum workflows and advanced DSP tasks: `dsp.SpectrumAnalyzer`.
Control System Toolbox	**Required**	Transfer-function and response analysis: `bode`, `tf`, `step`.
Audio Toolbox	**Required**	Audio-oriented sampling/analysis extensions used in later labs.
Simulink	**Required**	Model-based signal and control workflows in upcoming labs.

```
ver
license('test','Instrument_Toolbox')
license('test','Signal_Toolbox')
license('test','DSP_System_Toolbox')
license('test','Control_Toolbox')
license('test','Audio_Toolbox')
license('test','Simulink')
```

Deliverable (1/3): Upload a screenshot of MATLAB Add-On Manager showing Instrument Control, Signal Processing, DSP System, Control System, Audio Toolbox, and Simulink installed.

4.2 Arduino IDE 2.x & Library Setup

Do **not** use legacy Arduino IDE 1.x. Install **Arduino IDE 2.3+** from https://arduino.cc/en/software.

Table 2. Board Support Package (Boards Manager)

Package	Version	Purpose
Arduino AVR Boards	1.8.6+	Required for Arduino Mega 2560 target support.

Install path:

- Boards package: Tools → Board → Boards Manager → search `Arduino AVR Boards`.
- TimerOne: Tools → Manage Libraries → search `TimerOne` → install.

Table 3. Library Manager Requirements

Library	Status	Purpose
TimerOne (Jesse Tane)	Recommended (install 1.1.0+)	Simplifies Timer1 configuration for high-frequency PWM workflows in Lab 1 and beyond.

4.3 USB Drivers & Port Permissions

Driver issues are the most common cause of failed serial communication.

Table 4. USB Driver by Board Variant

Board Variant	USB Chip	Driver Requirement
Official Arduino Mega 2560	ATmega16U2	Native CDC-ACM support on modern Windows/macOS/Linux (usually no manual install).
Most clone/off-brand Mega boards	CH340G	Manual CH340 driver install required.
Some clone boards	FTDI FT232	Usually auto-installed; FTDI vendor driver may improve stability.

Table 5. CH340 Driver Installation Notes

OS	Source	Notes
Windows 10/11	https://www.wch-ic.com	Run installer as Administrator and reboot.
macOS Ventura/Sonoma+	https://www.wch-ic.com	Approve kernel/security prompt in System Settings → Privacy & Security.
Linux (kernel 3.12+)	Built-in	Run `sudo usermod -aG dialout $USER`, then log out/in.

4.4 Finding the Correct Serial Port

4.5 Oscilloscope Screenshot Export (Optional)

If your scope supports PC transfer, you may use vendor software (e.g., Tektronix OpenChoice Desktop or equivalent) to export PNG/PDF screenshots for reports. A USB thumb drive save from the front panel is also acceptable.

4.6 Installation Checklist

Complete the following checklist before attending Lab 1.

Table 6. Port Discovery by Operating System

OS	Where to Check	Typical Port Name
Windows	Device Manager → Ports (COM & LPT)	COM3, COM4, ...
macOS	Terminal: `ls /dev/tty.*`	`/dev/tty.usbmodem*`, `/dev/tty.wchusbserial*`
Linux	Terminal: `ls /dev/ttyUSB* /dev/ttyACM*`	`/dev/ttyUSB0`, `/dev/ttyACM0`

- ☐ MATLAB R2020a or later installed (R2023b+ recommended).
- ☐ Instrument Control Toolbox installed and licensed.
- ☐ Signal Processing Toolbox installed and licensed.
- ☐ DSP System Toolbox installed and licensed.
- ☐ Control System Toolbox installed and licensed.
- ☐ Audio Toolbox installed and licensed.
- ☐ Simulink installed and licensed.
- ☐ Arduino IDE 2.x installed.
- ☐ Arduino AVR Boards package installed via Boards Manager.
- ☐ TimerOne library installed via Library Manager.
- ☐ Correct USB driver installed for your board (CH340 if clone).
- ☐ Port permissions configured (Linux dialout / macOS approval if needed).
- ☐ Port name identified and verified in MATLAB script settings.
- ☐ Arduino Serial Monitor tested at 115200 baud.
- ☐ Arduino IDE Serial Monitor **closed** before running MATLAB (single-app COM port access).

Checkpoint (1/1): Show your TA your completed MATLAB/Arduino installation and the functional Checklist.

Deliverable (2/3): Upload screenshots of: (1) MATLAB `ver` output, (2) toolbox license checks, and (3) Arduino IDE Boards/Library Manager pages showing AVR Boards and TimerOne.

4.7 MATLAB Tutorial

Follow the MATLAB Onramp tutorial: https://www.mathworks.com/learn/tutorials/MATLAB-onramp.html.

Deliverable (3/3): Upload your completed MATLAB Onramp Certificate PDF.

5 Laboratory Assignment

Submit the following to Gradescope:

- **Pre-lab Deliverable 1:** Lab Door Selfie.
- **Deliverable 1:** MATLAB Add-On Manager screenshot with all required toolboxes.
- **Deliverable 2:** MATLAB `ver` + toolbox license-check screenshots.
- **Deliverable 3:** Arduino IDE Board/Library Manager screenshots (AVR Boards + TimerOne).
- **Deliverable 4:** MATLAB Onramp Certificate PDF.

6 Updates

- April 2026: Updated for MATLAB R2023b, Arduino IDE 2.x, and CH340 driver requirements.
- January 2025: Updated lab location and enumerated add-ons.